

Coding & Solution

Date	30 June 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Marehemalatha/Comprehensive-measure-of-well-being-codefiles
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	• HDI prediction using Linear Regression • Dataset preprocessing • Data visualization using Matplotlib and Seaborn • Missing value handling • Train-Test Split • Model training and prediction • Save trained model using Pickle • Flask web application • User-friendly prediction interface
Pending / Incomplete Features	• Deployment to cloud (optional)

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	No
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

This project predicts the Human Development Index (HDI) score using a Linear Regression machine learning model. The application is developed using Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, and Flask. The trained model is integrated with a web interface that allows users to enter socio-economic indicators and obtain the predicted HDI category.